CLASS – TYBBA CA , TY BCA

ASSIGNMENT – 3

SUBJECT – ESSENTIALS OF CLOUD COMPUTING

SECTION (A) - Answer the following questions in short: (2/3 Marks each)

Q1. Differentiate between block storage, file storage, and object storage in terms of their data unit and access method.

Q2.State the CAP theorem and explain why partition tolerance is generally considered non-negotiable in practice.

Q3. Distinguish between durability and availability in the context of cloud storage.

Q4. What is the difference between synchronous and asynchronous replication?

SECTION (B) - Answer the following questions in detail: (10 Marks each)

Q5. Explain the three fundamental cloud storage architectures (block, file, and object) in detail, including their key characteristics, typical use cases, and cloud provider examples. Conclude with a comparison of when each would be the most appropriate choice.

Q6. Describe the CAP theorem in detail, including the meaning of Consistency, Availability, and Partition Tolerance. Discuss the CP vs. AP trade-off with examples and explain how the PACELC extension builds on this theorem for normal (non-partitioned) operating conditions.

Q7. Discuss data replication and redundancy in cloud storage. Explain the different types of replication (synchronous, asynchronous, semi-synchronous) and different replication scopes (local, zone-redundant, region-redundant), and explain why replication alone is not sufficient for a complete data protection strategy.

Q8. Compare and contrast SQL and NoSQL databases in terms of schema, scaling approach, consistency model, and ideal workloads. Describe at least three types of NoSQL databases with examples, and explain the role of NewSQL as a hybrid approach.